

Liam Tanner

250-510-5933 | liam.tanner@brentwood.ca | [linkedin.com/liam-tanner](https://www.linkedin.com/liam-tanner) | github.com/liamtannerr

TECHNICAL SKILLS

Programming Languages: Python, C++, C, JavaScript, Linux Bash
Infrastructure & CI/CD: Docker, PostgreSQL, MongoDB, Git, GitHub Actions, Automated Testing
AI & Machine Learning: PyTorch, LLM Orchestration, RAG, Convolutional Neural Networks
Backend Architecture: Distributed Systems, REST APIs (FastAPI), Object-Oriented Design, Hardware Optimization

EDUCATION

University of Victoria Victoria, BC
Bachelor of Software Engineering Sept. 2021 – Aug 2026 (expected)

- 4.00/4.00 GPA – 93% Average (2025/2026 Session)
- Relevant Coursework: Operating Systems, Databases, Communication Networks, Optimization for Machine Learning

EXPERIENCE

Software Engineer Jan 2025 – Present
AUVIC - Autonomous Underwater Vehicles Club Victoria, BC

- Represented the University of Victoria at RoboSub 2026 in Irvine, California, resolving critical thruster saturation limits in the C++ control loop and engineering a state machine for autonomous navigation, advancing the team to the semi-finals of an international engineering competition.
- Engineered a ROS 2 dynamic configuration pipeline to extract PID and Kalman filter values from YAML files, eliminating C++ recompilation downtime and enabling real-time parameter tuning during pool testing.

Software Developer June – Dec 2025
Natural Resources Canada Victoria, BC

- Collaborated in an Agile Scrum environment to design and implement a scalable data transformation workflow, processing massive PostgreSQL datasets and reducing execution times from hours to seconds.
- Engineered an encrypted database release-versioning system using SQL to securely track metadata, ensuring full-lifecycle accountability and seamless feature rollouts across 2-week development sprints.
- Rebuilt a legacy frontend infrastructure from scratch in JavaScript, writing custom Bash scripts to automate the entire process for all engineers on my team.

Research Assistant - Induced Seismicity Project Sept 2023 – April 2024
Natural Resources Canada Sidney, BC

- Engineered 3 increasingly complex CO₂ injection simulations using Python meshes and the OpenGeoSys finite element solver, becoming the first NRCan employee to successfully implement this modeling workflow.
- Advanced the project to a real-world case study by building a 3-layer finite element simulation of the SECURE INGA wastewater disposal well, utilizing precise depth profiles and material properties.

PROJECTS

AI Incident Orchestrator | *Python, PostgreSQL, Groq API, Docker* May 2026 – Present

- Engineered a containerized AI orchestration pipeline using Docker Compose (FastAPI, React, Nginx) to automate suggested action mapping of cybersecurity threats via LLMs and RAG.
- Streamlined CI/CD rollouts for a 17-member team by automating testing via GitHub Actions and engineering a custom Bash/GitLab deployment pipeline.

Convolutional Engine 🌀 | *CUDA, C++, Hardware Optimization* Apr 2026

- Built a high-performance image processing library from scratch; implemented shared memory tiling algorithms to bypass global memory bottlenecks and maximize GPU thread throughput.

EfficientAD Anomaly Detector 🌀 | *PyTorch, CNN Architecture* Mar 2026

- Engineered a state-of-the-art PyTorch anomaly detection pipeline by writing a Patch Description Network, training the teacher network, and removing autoencoder bottlenecks to run 70,000 training iterations in under 2 hours, achieving an 0.88 AUROC with sub-millisecond performance, allowing for real-time inference

Virtual Triage System 🌀 | *Python, FastAPI, MongoDB, ReactJS* Sept – Nov 2024

- Engineered an emergency department, virtual triaging web application, using a Python backend and ReactJS frontend, that allows patients to wait for an available physician from home.